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Seoul, S. Korea

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Google Scholar

Minwoo Jung

EDUCATION

Seoul National University, Republic of Korea *Sep. 2023 - Present*

- **Ph.D. student** in Mechanical Enginnering
- Advised by Dr. Ayoung Kim

Seoul National University, Republic of Korea *Aug. 2023*

- **M.S. student** in Mechanical Enginnering
- Cumulative GPA 4.06/4.3
- Dissertation: "Asynchronous Multiple LiDAR-Inertial Odometry using Point-wise Inter-LiDAR Uncertainty Propagation"
- Advised by Dr. Ayoung Kim

KAIST, Republic of Korea *Mar. 2021 - Aug. 2021*

- **M.S. student** in Civil and Environmental Engineering
- Moved to Seoul National University
- Advised by Dr. Ayoung Kim

KAIST, Republic of Korea *Mar. 2017 - Feb. 2021*

- B.S. in Civil and Environmental Engineering
- Cumulative GPA 3.76/4.3
- Graduated *cum laude*

LANGUAGES & SKILLS

- Korean (native), English (slightly fluent)
- MATLAB, C/C++, Python, ROS, Pytorch, Docker
- LaTeX, Microsoft Office, Ubuntu, Windows

RESEARCH INTEREST

- Multiple-LiDAR-Inertial Odometry
- Sensor Fusion for SLAM
- LiDAR Place Recognition

RESEARCH EXPERIENCE

Autonomous Navigation in Construction Sites *2021 - Present*

- Graduate Student Research Assitant (RPM Robotics Lab)
- Implemented autonomous navigation module for a mobile robot equipped with LiDAR, IMU and GPS, demonstrated across multiple construction sites.

Global Re-localization for LiDAR SLAM *Sep. 2024*

- Graduate Student Research Assitant (RPM Robotics Lab)
- Developed global re-localization for localizing a robot on a prior map in GPS-denied environments through scan-to-submap registration

- Integrated the global re-localization module into the existing SLAM framework and validated it using real-word multi-session data.

Multiple LiDAR-IMU SLAM *2022 - 2023*

- Graduate Student Research Assitant (RPM Robotics Lab)
- Developed the Multiple LiDAR-IMU SLAM framework to estimate the trajectory in unstructured environments.
- Integrated the module into the hardware platform and validated it using real-word forest.

PUBLICATIONS

International Journal

1. Jeongyun Kim, Myung-Hwan Jeon, Sangwoo Jung, Wooseong Yang, Jung, Minwoo, Jaeho Shin, and Ayoung Kim. Transpose: Large-scale multispectral dataset for transparent object. *The International Journal of Robotics Research*, page 02783649231213117, 2024
2. Sangwoo Jung, Hyesu Jang, Jung, Minwoo, Ayoung Kim, and Myung-Hwan Jeon. Imaging radar and lidar image translation for 3-dof extrinsic calibration. *Intelligent Service Robotics*, 17(2):167–179, 2024
3. Dongjae Lee, Jung, Minwoo, Wooseong Yang, and Ayoung Kim. Lidar odometry survey: recent advancements and remaining challenges. *Intelligent Service Robotics*, 17(2):95–118, 2024
4. Hyesu Jang, Jung, Minwoo, Myung-Hwan Jeon, and Ayoung Kim. Lodestar: Maritime radar descriptor for semi-direct radar odometry. *IEEE Robotics and Automation Letters*, 2024
5. Jung, Minwoo, Wooseong Yang, Dongjae Lee, Hyeonjae Gil, Giseop Kim, and Ayoung Kim. Helipr: Heterogeneous lidar dataset for inter-lidar place recognition under spatiotemporal variations. *The International Journal of Robotics Research*, 43(12):1867–1883, 2024
6. Jung, Minwoo, Sangwoo Jung, and Ayoung Kim. Asynchronous multiple lidar-inertial odometry using point-wise inter-lidar uncertainty propagation. *IEEE Robotics and Automation Letters*, 8(7):4211–4218, 2023
7. Jeongyun Kim, Seungsang Yun, Jung, Minwoo, Ayoung Kim, and Younggun Cho. Automatic wall slant angle map generation using 3d point clouds. *ETRI Journal*, 43(4):594–602, 2021

Domestic Journal

1. Sangwoo Jung, Jung, Minwoo, and Ayoung Kim. Map error measuring mechanism design and algorithm robust to lidar sparsity. *The Journal of Korea Robotics Society*, 16(3):189–198, 2021
2. Jung, Minwoo, Sangwoo Jung, Hyesu Jang, and Ayoung Kim. Intensity and ambient enhanced lidar-inertial slam for unstructured construction environment. *The Journal of Korea Robotics Society*, 16(3):179–188, 2021

International Conference Proceedings

1. Sanghyun Hahn, Seunghun Oh, Jung, Minwoo, Ayoung Kim, and Sangwoo Jung. Quantitative 3d map accuracy evaluation hardware and algorithm for lidar (-inertial) slam. *arXiv preprint arXiv:2408.09727*, 2024 (ICCAS accepted)
2. Hyesu Jang, Jung, Minwoo, and Ayoung Kim. Raplace: Place recognition for imaging radar using radon transform and mutable threshold. In *2023 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pages 11194–11201. IEEE, 2023
3. Seungsang Yun, Jung, Minwoo, Jeongyun Kim, Sangwoo Jung, Younghun Cho, Myung-Hwan Jeon, Giseop Kim, and Ayoung Kim. Sthereo: Stereo thermal dataset for research in odometry and mapping. In *2022 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pages 3857–3864. IEEE, 2022

Workshop

1. Dongjae Lee, Jung, Minwoo, and Ayoung Kim. Conpr: Ongoing construction site dataset for place recognition. *arXiv preprint arXiv:2407.03684*, 2024 (Field Robotics Workshop in ICRA 2024)

2. Eunho Lee, Jung, Minwoo, and Ayoung Kim. Toward robust lidar based 3d object detection via density-aware adaptive thresholding. *arXiv preprint arXiv:2404.13852*, 2024 (Closing the Loop on Localization Workshop in IROS 2023)
3. Hyesu Jang, Jung, Minwoo, Sangwoo Jung, and Ayoung Kim. Radar-based angular correction for tightly-coupled lidar odometry (PNARUDE Workshop in IROS 2022)

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2. Hyesu Jang, Jung, Minwoo, and Ayoung Kim. Raplace: Place recognition for imaging radar using radon transform and mutable threshold. In *2023 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pages 11194–11201. IEEE, 2023
3. Seungsang Yun, Jung, Minwoo, Jeongyun Kim, Sangwoo Jung, Younghun Cho, Myung-Hwan Jeon, Giseop Kim, and Ayoung Kim. Sthereo: Stereo thermal dataset for research in odometry and mapping. In *2022 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pages 3857–3864. IEEE, 2022

Workshop

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3. Hyesu Jang, Jung, Minwoo, Sangwoo Jung, and Ayoung Kim. Radar-based angular correction for tightly-coupled lidar odometry (PNARUDE Workshop in IROS 2022)

REVIEWER

- The International Journal of Robotics Research (IJRR)
- The International Conference on Control, Automation and Systems (ICCAS)
- The International Journal of Digital Earth (IJDE)
- IEEE Robotics and Automation Letters (RA-L)
- IEEE International Conference on Intelligent Robots and Systems (IROS)
- IEEE Transactions on Automation Science and Engineering (T-ASE)

AWARDS AND HONORS

2023	“Best Overall Presentation Award”, IROS Workshop
2023	“Seoul National University Best (M.S) Thesis Award”, SNU
2023	“Ranked 4th in the Lidar Single Session category”, Hilti SLAM Challenge
2021	“Graduated Citation”, the Korea Concrete Institute
2021	“Cum Laude Grades”, CEE, KAIST
2019	“Dean’s list”, CEE, KAIST